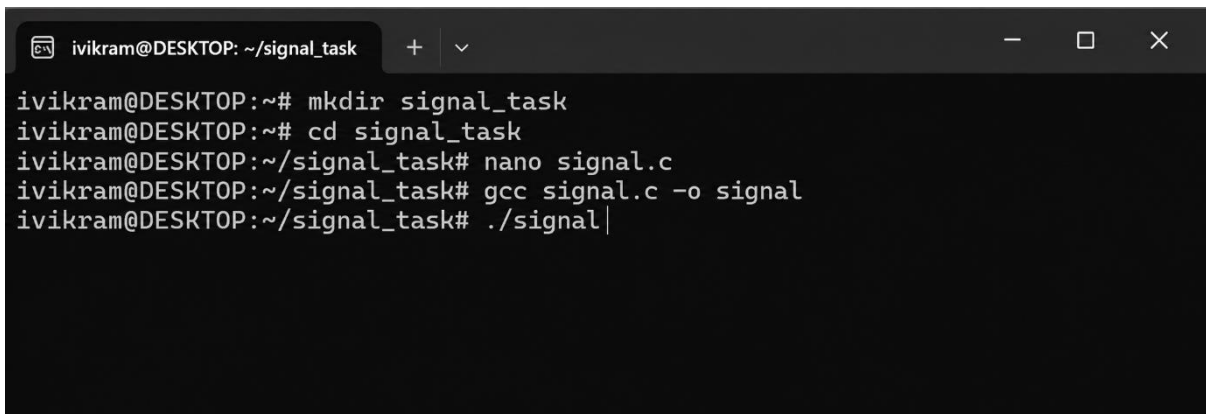


Practical-6

Task - 2:

Create a POSIX signal handling program that captures SIGINT, SIGTERM, and SIGUSR1. Demonstrate asynchronous event handling and explain the role of signal handlers.



```
ivikram@DESKTOP: ~/signal_task
ivikram@DESKTOP:~# mkdir signal_task
ivikram@DESKTOP:~# cd signal_task
ivikram@DESKTOP:~/signal_task# nano signal.c
ivikram@DESKTOP:~/signal_task# gcc signal.c -o signal
ivikram@DESKTOP:~/signal_task# ./signal|
```

Code:

```
#include <stdio.h>

#include <stdlib.h>

#include <unistd.h> #include <signal.h> void handle_signal(int sig) { if (sig ==
SIGINT) { printf("\nReceived SIGINT (Ctrl+C)\n"); printf("Program is still
running...\n");
}
else if (sig == SIGTERM) { printf("\nReceived SIGTERM\n");
printf("Program terminating...\n"); exit(0);
}
```

```

    else if (sig == SIGUSR1) {        printf("\nReceived
SIGUSR1\n");        printf("Asynchronous event handled
successfully.\n");
    }
}

int main() {    signal(SIGINT, handle_signal);    signal(SIGTERM,
handle_signal);    signal(SIGUSR1, handle_signal);
printf("POSIX Signal Handling Program\n");    printf("Process ID:
%d\n", getpid());    printf("Press Ctrl+C to generate SIGINT.\n");
printf("Use kill -USR1 %d to generate SIGUSR1.\n", getpid());
printf("Use kill -TERM %d to generate SIGTERM.\n\n", getpid());

    while (1) {        printf("Program is
running...\n");        sleep(3);
    }

    return 0;
}

```

Output:

```
ivikram@DESKTOP: ~/signal_task X + v
ivikram@DESKTOP:~# mkdir signal_task
ivikram@DESKTOP:~# cd signal_task
ivikram@DESKTOP:~/signal_task# nano signal.c
ivikram@DESKTOP:~/signal_task# gcc signal.c -o signal
ivikram@DESKTOP:~/signal_task# ./signal
POSIX Signal Handling Program
Process ID: 445
Press Ctrl+C to generate SIGINT.
Use kill -USR1 445 to generate SIGUSR1.
Use kill -TERM 445 to generate SIGTERM.

Program is running...
Program is running...
Program is running...
Program is running...
^C
Received SIGINT (Ctrl+C)
Program is still running...
Program is running...
Program is running...
Program is running...
Program is running...
|
```

```
ivikram@DESKTOP: ~/signal_task X + v
ivikram@DESKTOP:~/signal_task# ./signal
POSIX Signal Handling Program
Process ID: 515
Press Ctrl+C to generate SIGINT.
Use kill -USR1 515 to generate SIGUSR1.
Use kill -TERM 515 to generate SIGTERM.

Program is running...
4Program is running...
5Program is running...
^C
Received SIGINT (Ctrl+C)
Program is still running...
Program is running...
445
Program is running...
Program is running...
Program is running...
Program is running...

ivikram@DESKTOP: ~ X + v
ivikram@DESKTOP:~# kill -USR1 445
ivikram@DESKTOP:~# kill -TERM 445
ivikram@DESKTOP:~# kill -USR1 445
-bash: kill: (445) - No such process
ivikram@DESKTOP:~#
```

OBSERVATION: The program successfully captured and handled the SIGINT, SIGTERM, and SIGUSR1 signals using POSIX signal handling. A signal handler was registered for each signal, and when a signal was sent to the running process, the corresponding handler was executed and the appropriate message was displayed. The experiment demonstrated asynchronous event handling, where signals can be received by a process at any time during its execution. Thus, the role of signal

handlers in responding to and managing asynchronous events was successfully demonstrated.